

DSST389: Advanced Data Science

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Course Notes



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Overview

Introduction

Sample text.¹

Anthology of Computers and the Humanities, Vol. 3

Sitcom Form and Function: Pacing and Production in a Collection of Thirty U.S. Series

Taylor Arnold¹ and Lauren Tilton²

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Abstract

Using an original corpus of thirty U.S. situational comedies (sitcoms) spanning over 4,500 episodes, we investigate how visual and aural pacing evolve over time and across modes of production. We find a clear trend toward faster pacing in visual editing, spoken dialogue, and textual density throughout the decades. While this shift correlates strongly with changes between multi-camera and single-camera setups, it is also shaped by the narrative goals of each series. For example, *Seinfeld* and *Friends*, despite sharing visual and production similarities with other 1990s multi-camera sitcoms, feature markedly faster dialogue that reflects a narrative emphasis on wit and language. In contrast, single-camera series such as *Modern Family* and *The Office (US)* combine rapid dialogue with long takes and visual pauses that support physical and situational humor.

By combining large-scale computational analysis with close attention to aesthetic and narrative function, this study contributes to ongoing debates in television theory regarding the relationship between form and meaning. Whereas some scholars have emphasized the sitcom's formal conservatism and narrative stability (Mills; Ellis), our findings reveal a more dynamic interplay between production technology, pacing, and storytelling strategy. Drawing from media-specific approaches (Caldwell; Jacobs) and cultural theory, we argue that sitcom style emerges through a negotiation between material affordances and discursive intentions. This approach reframes how we understand the evolution of sitcom aesthetics and offers new empirical insight into the genre's formal diversity and cultural significance.

Keywords: computational television studies, multimodal analysis, situational comedies, distant viewing, digital humanities

Front page our CHR paper, with the initial results.

¹ Arnold and Tilton 2025.



Overview

Code Example

Code slides use a wider right column so a full 72-character line of code fits without overflowing.

The left column is for narration.

```
import polars as pl
from plotnine import ggplot, aes
from polars import col as c

import funs

country = pl.read_csv("data/countries.csv")
```



References

Arnold, Taylor and Lauren Tilton (2025). "Sitcom Form and Function: Pacing and Production in a Collection of Thirty U.S. Series". In: *Computational Humanities Research 2025*. Ed. by Taylor Arnold, Margherita Fantoli, and Ruben Ros. Anthology of Computers and the Humanities, pp. 249–265. DOI: [10.63744/yHo626es4FhQ](https://doi.org/10.63744/yHo626es4FhQ).